LaTeX Workshop

Physics and Astronomy Society

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Overview

1. Set up
2. Initial set up of Documents
3. Formatting Content
4. Images, Lists and Tables
5. Maths
6. Referencing
7. Error Handling

Latex Editor

- Latex is a 'markup language' you write in.
- It requires an editor, such as **Overleaf**
- Think Python $\longrightarrow$ Visual Studio

- Downloadable editors:
- MikTex $\longrightarrow$ TeXworks



Overleaf is the most widespread editor as it is cloud based and user friendly

Starting a Document

- The most basic three things:
- A document class
- A beginning
- An ending

```
1
2  \documentclass{article}
3  \begin{document}
4  Hello World !!! % your content
5  \end{document}
```

How to Set up a Document

Latex Documents are split into 2 sections Preamble and Content:
Preamble for initial set up:

- Enter Document Class
- Initialize Packages
- Define Page Geometry
- Define Command (If your a lunatic)

Content is where all your text will go and it is what will be outputted by the compiler

```
1  %Preamble
2  \documentclass{article}
3  \usepackage{graphicx}
4  \usepackage{amsmath}
5  \usepackage{amssymb}
6
7  \begin{document}
8  % Content
9  \end{document}
```

Formatting Content

LaTeXfollows a sectioning order that allows you to partition your documents properly.

1. Part
2. Chapter
3. Section
4. Subsection
5. Subsubsection
6. Paragraph

Unless your writing a book or a research paper the only partitions you need are:
Section, Subsection and Subsubsection as they are automatically compatible with a great command called:
tableofcontents
This command auto generates a table of contents with the sections

Formatting Content

Often you will want to format your text in particular ways i.e.

- **Bold Text**
- *Italics*
- Font Changes
- HUGE TEXT
- Small Capitals

Similarly you might need to align your text

Maybe You want it to be center aligned

Maybe you want it left aligned

How about a nice justified text

Or even Right Aligned

Loading Images & using Lists and Tables

```
\begin{figure}
    \centering
    \includegraphics[width=0.5\linewidth]{Example.png}
    \caption{Caption}
    \label{fig:enter-label}
\end{figure}
```

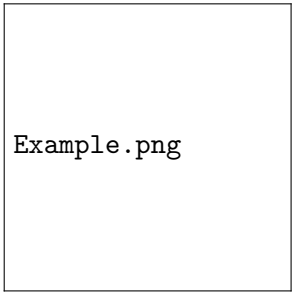


Figure: Caption

Loading Images & Using Lists and Tables

```
\begin{table}[]
    \centering
    \begin{tabular}{|c|c|} \hline
        Temperature $(^\circ C)$ & Density ($g$ $cm^{-3}
        \hline
         18& 9.996 \\ \hline
    \end{tabular}
    \caption{Caption}
    \label{tab:my_label}
\end{table}
```

Temperature $(^\circ C)$	Density $(g\ cm^{-3})$
18	9.996

Table 1: Caption

Using Maths

Using LaTex Maths is usually the main reason anyone uses LaTeX to write up documents in science
LaTex has two methods to write math:
You can do inline math which inserts equations in a sentence $x^2 + y^2 = r^2$ like this.
But there is also a maths mode that center aligns your math and numbers equations to allow you to reference it.

$$\int_{-\infty}^{\infty} |f(x)|^2 dx < \infty \longrightarrow \therefore f(x) \in \mathcal{L}^2 \ \ \& \ \ f(x) \in \mathcal{H} \tag{1}$$

Please see eqn.1 on what a center aligned and referenced equation looks like

Referencing

Referencing can be incredible useful in documents to refer back to previously established material: Latex has an automatic referencing system:
i.e Latex remembers the number of your table and will insert it for you:

It uses two commands label{} and ref{}
Insert label command inside your begin{} % content end{} chain (See Slide 9)
and use ref{Name of Label} where you want the number to appear.

Error Handling

There is two main ways to fix errors in Latex:

1. Read the Error Logs
2. Google it

There is a Massive Latex Stack exchange someone has had your problem before.

